Ignition Heatmaps from SEVIRI Fire Radiative Power

Master thesis — progress update and open points

Rafael Gufler

October 2026

In one slide

Task

Input: 24 h of SEVIRI FRP hotspots (15-min scans, $\approx 3 \times 4.5$ km pixels), Portugal 2004–2019.
Output: an **ignition probability map** — where did fires ≥ 50 ha start in the middle 12 h?
Ground truth: ICNF fire register (ignition point, start time, burned area).

Where it stands

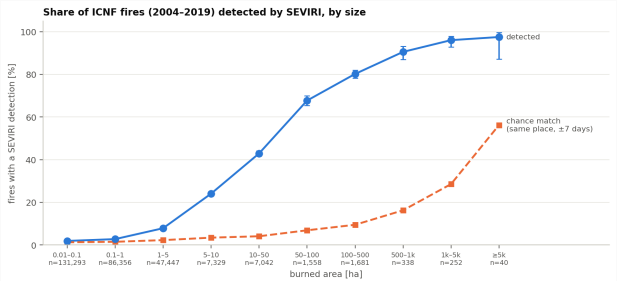
- Data pipeline, baseline, two models done
- Test 2017–19 evaluated **once**
- AP@5 km **0.62** vs. 0.50 baseline
- AP@10 km **0.71** vs. 0.59 baseline

To discuss today

- The heatmap framing and evaluation
- Data: SEVIRI 2020+, cloud mask, rekindles
- Priorities: covariates, Transformer, space–time output, explainability

Data

The data: what SEVIRI can see



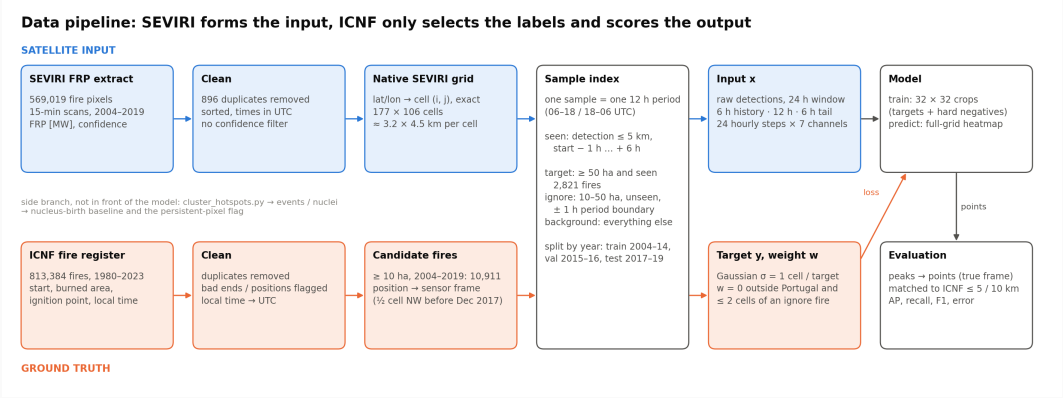
- **SEVIRI**: 568k hotspot detections (pixel $\times$ 15-min slot), FRP + confidence; only fire pixels stored
- **ICNF**: 361k fires with coordinates in 2004–2019; only 2,311 $\geq$ 100 ha
- Detected: 43 % at 10–50 ha, 68 % at 50–100 ha, $\geq$ 80 % from 100 ha

$\Rightarrow$ **Targets: fires $\geq$ 50 ha that SEVIRI sees** (2,821)

Cleaning findings

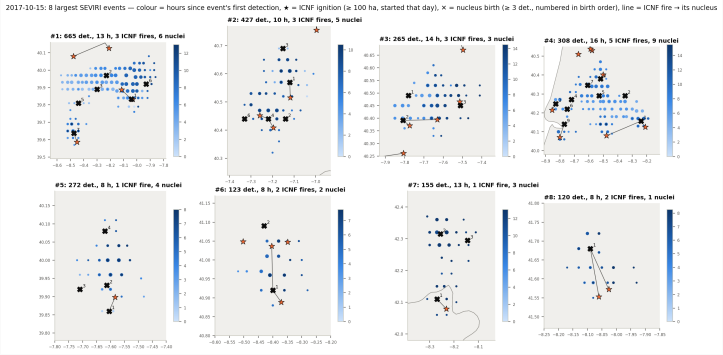
ICNF times are **local time**, not UTC (lag jumps 0.9 h at the DST switch).
No confidence filter: it removes the weak, early detections.

Data pipeline



Native SEVIRI grid recovered exactly (177 × 106 cells, no resampling) · chronological split: train 2004–14 / val 2015–16 / **test 2017–19 locked**

The core difficulty: simultaneous neighbouring fires



2017-10-15: SEVIRI events, ICNF ignitions (⋆), nucleus births (×)

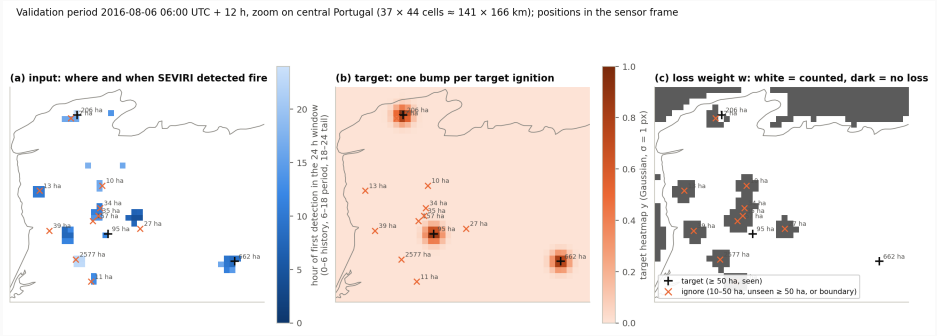
- 1 in 4 large fires has another ≥ 100 ha fire within 10 km on the same day
- 1 in 3 shares $> 50\,\%$ of its hotspots with another fire
- Space–time clustering (6 km, 6 h) into events and **nuclei**: works for isolated fires, fails next to active fires

Use of the clustering

Baseline: every nucleus birth = predicted ignition, ranked by its fire radiative energy.

Approach

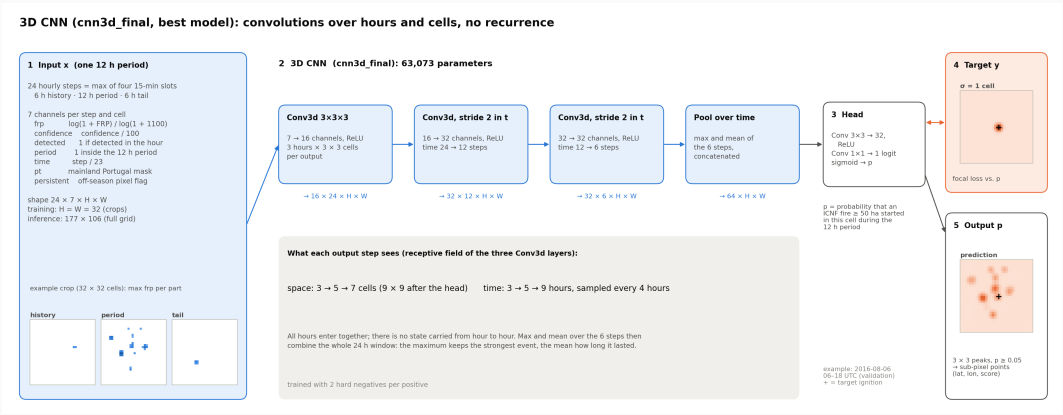
Framing: ignition heatmaps from raw hotspots



- No hand-made hotspot ↔ fire assignment: the model learns it
- One sample = one 12 h period; input 6 h history + 12 h + 6 h tail, hourly

- Target: Gaussian ($\sigma = 1$ cell) at *every* ignition $\Rightarrow$ 0, 1 or 31 per map
- Ignore regions: 10–50 ha and unseen fires, Spain

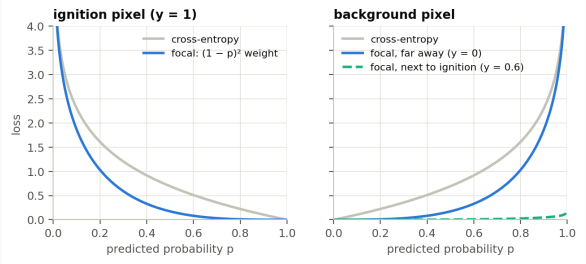
Model: 3D CNN (main) and ConvLSTM (proposal)



Training: 32 × 32 crops; one per target + **2 hard negatives** (fire present, no new ignition); flips; 3 seeds.

ConvLSTM: same input and head, 2 recurrent layers × 16 channels (≈45k params). Equal on test.

Loss and evaluation



Focal loss (CornerNet / CenterNet), $\alpha = 2,\ \beta = 4$

ignition pixels: $-(1-p)^{\alpha} \log p$
other pixels: $-(1-y)^{\beta}\, p^{\alpha} \log(1-p)$

easy pixels down-weighted · near-misses cheap · normalised per ignition

From heatmap to points

3×3 local maxima, $p \geq 0.05$, sub-pixel centroid $\rightarrow$ lat/lon

Matching and metrics

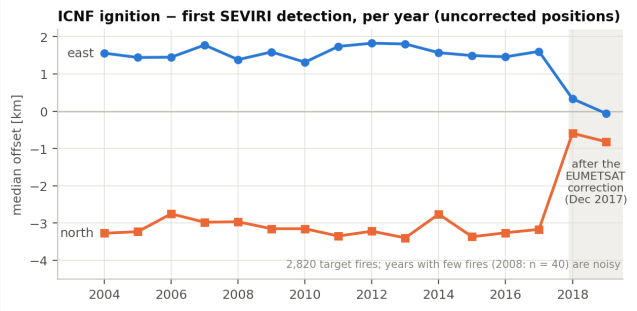
Per period, by score: nearest unmatched ICNF ignition within k = hit; near an ignore fire = neither; else false alarm.

$k = 5$ km (location) and 10 km (right fire).

AP = area under precision–recall: how well scores rank real ignitions, over all thresholds.

Results

Finding: a half-pixel geo-referencing offset

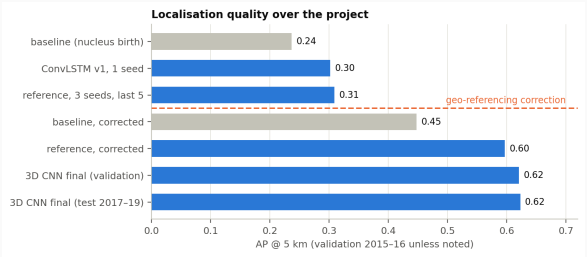


- Symptom: *no flip augmentation* almost doubled AP@5 km ⇒ a directional pattern in the data
- Step change in Dec 2017 of **half a pixel** north-west, as documented for SEVIRI L1.5 (satpy / EUMETSAT)
- Fix: labels moved into the sensor frame before 2017-12-01

Effect

Baseline error 4.3 → 2.2 km; model AP@5 km 0.35 → 0.60; seed spread ±0.05 → ±0.016.

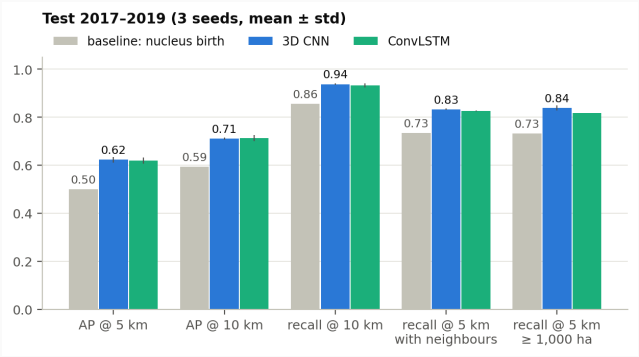
Progress and ablations (3 seeds each)



Question	Answer
Recurrence needed?	No: 3D CNN $\geq$ ConvLSTM
Bigger model?	No: 16 ch $\geq$ 32, 1 layer $\approx$ 2
FRP intensity?	Little: presence only -0.01
6 h history?	Small gain ($+0.02$)
Clustering as input?	No gain
Hard negatives?	$\times 2$: $+0.015$
15-min steps?	No gain at $4\times$ cost

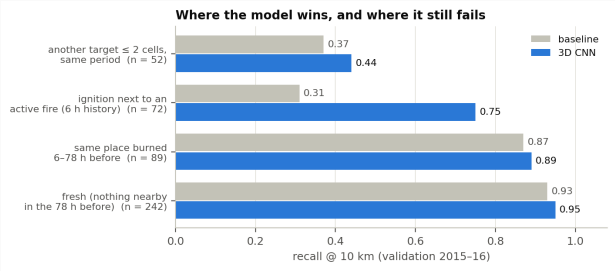
All variants within ± 0.03 AP $\Rightarrow$ **limited by labels and data, not by architecture.**

Test results 2017–2019 (evaluated once)



- +0.12 AP at both radii; fewer false alarms (0.10 vs. 0.15 per day)
- Validation numbers replicate on test (0.635 / 0.704 vs. 0.623 / 0.711)
- 3D CNN ≡ ConvLSTM
- Largest gains for fires with neighbours and large fires
- Median error ≈ 2.0 km for both model and baseline: the gain is in *finding and ranking*, not sub-pixel precision

Where it wins and where it fails



- **Win**: ignitions next to active fires ($0.31 \rightarrow 0.75$)
- **Fail**: two ignitions ≤ 2 cells apart in one period ($\approx$70 % of misses); typically $\sim$1.7 h apart $\Rightarrow$ a space–*time* output could separate them
- 58 % of "false alarms" sit on real fires (< 10 ha, or started 1–72 h before)
- Localisation floor ≈ 1.6 km (nearest detected pixel); model 1.9–2.0 km

Discussion

Open points for discussion

Decisions to confirm

- Heatmap framing, ≥ 50 ha, "seen" fires only
- Early detection vs. post-fire reconstruction
- AP@5/10 km as main metrics
- Geo-offset: own section? exact EUMETSAT date?

Data requests

- SEVIRI FRP 2020–2024 (LSA SAF)
- SEVIRI cloud mask
- ICNF rekindle flag, burned-area perimeters
- ICNF ignition-point accuracy
- Full reference of "Xu et al. 2025"

Next steps — priority?

- Space–time ignition output
- Covariates (wind, DEM, FWI) — obj. iv
- Model 2: CNN + temporal attention (U-TAE)
- Explainability — obj. v
- DEM parallax test; leave-years-out CV

Caveats

Results conditional on SEVIRI seeing the fire (19 % of ≥ 50 ha unseen) · ICNF positions uncertain · test dominated by 2017 · improved decoder/ensemble not pre-registered · loss ablation only on uncorrected data · no cloud mask

Discussion

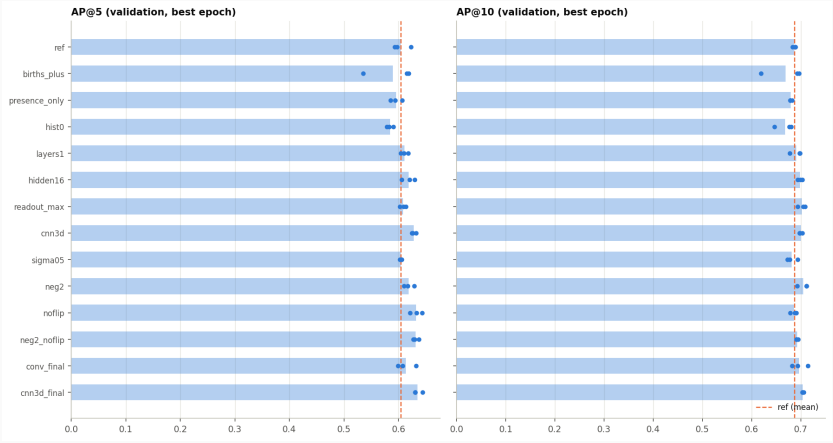
Ignition Heatmaps from SEVIRI Fire Radiative Power

Appendix

A1 — Status against the proposal objectives

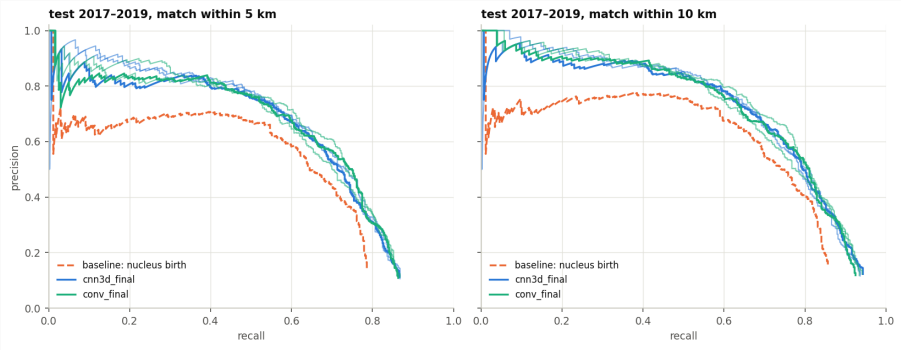
Objective	Status	Detail
(i) Dataset: FRP + environment	partial	FRP part done; covariates not yet
(ii) Design / evaluate architectures	partial	ConvLSTM + 3D CNN on test; Transformer open
(iii) Compare modelling strategies	largely done	raw vs. clustered, recurrent vs. conv., time steps, losses
(iv) Environmental variables	open	needs covariates
(v) Robustness, accuracy, interpretability	partial	subgroups and years done; XAI open
Extension to Europe	not started	optional

A2 — Ablations on the corrected data



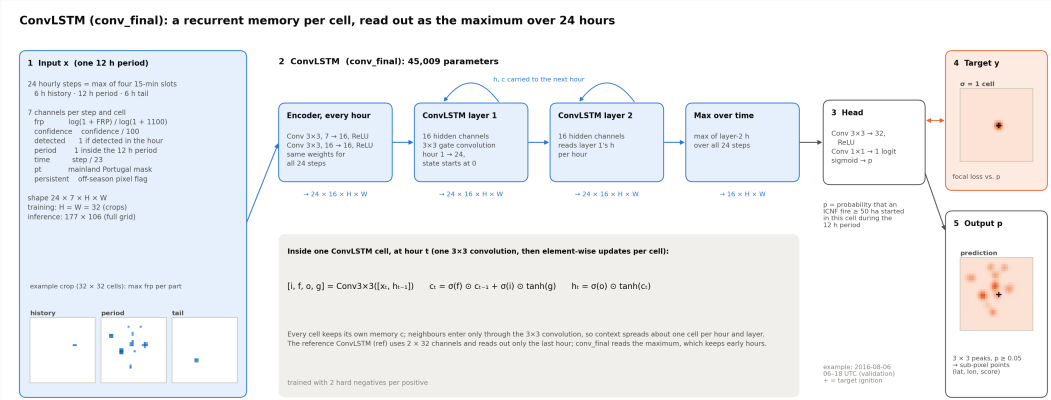
Validation 2015–16, best epoch; dots = seeds, dashed line = reference mean.

A3 — Precision–recall curves on test

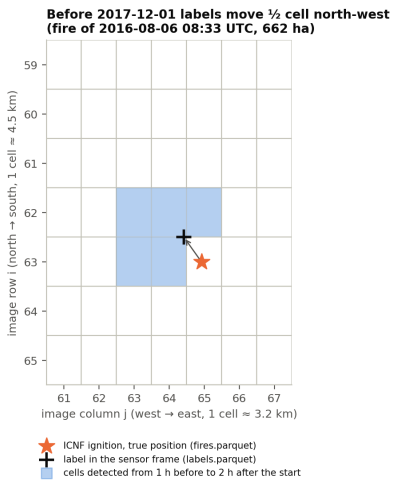


AP@5 / 10 km	2017	2018	2019
Baseline	0.45 / 0.55	0.71 / 0.77	0.68 / 0.75
3D CNN	0.58 / 0.68	0.85 / 0.85	0.73 / 0.80
ConvLSTM	0.59 / 0.69	0.86 / 0.86	0.68 / 0.77

A4 — ConvLSTM architecture



A5 — Geo-referencing correction for one fire



- Detections stay on their exact native pixels
- Labels before 2017-12-01 are moved half a cell north-west into the sensor frame
- Predictions are moved back before scoring against ICNF
- Remaining ≈ 0.2 cell southward offset in both periods, larger in mountain districts
  $\Rightarrow$ hypothesis: **terrain parallax** (satellite views from the south at $\approx 47^\circ$); testable with a DEM

A6 — Positioning in the literature

Four strands; none learns ignition location from geostationary FRP

- Geostationary detection/FRP (Wooster 2015; Koltunov 2016)
- Rule-based events and ignitions (Benali 2016; Andela 2019; Oom 2016; Chen 2022)
- DL on fire time series, mostly forward (Huot 2022; Gerard 2023); Pinto 2020 dates burning
- Generic: ConvLSTM, U-TAE, CenterNet / CornerNet / DARK

Closest connections

- **Benali 2016**: same country and ICNF; rule-based; $\sim$2 km / 12 h
- **Pinto 2020 (BA-Net)**: CNN + LSTM ablation, erase augmentation, per-pixel dating
- **OpenSTL (Tan 2023)**: recurrence-free $\approx$ ConvLSTM
- **Oom 2016 / FIRED**: cluster thresholds drive results
- **Paugam et al. 2026**: MTG-FCI tracking (successor sensor)

A7 — Key references

- Andela, N., et al. (2019). The Global Fire Atlas of individual fire size, duration, speed and direction. *Earth Syst. Sci. Data*, 11.
- Benali, A., et al. (2016). Determining fire dates and locating ignition points with satellite data. *Remote Sensing*, 8(4), 326.
- Chen, Y., et al. (2022). California wildfire spread derived using VIIRS satellite observations and an object-based tracking system. *Scientific Data*, 9.
- Garnot, V. S. F., & Landrieu, L. (2021). Panoptic segmentation of satellite image time series with convolutional temporal attention networks. *ICCV*.
- Law, H., & Deng, J. (2018). CornerNet: Detecting objects as paired keypoints. *ECCV*.
- Lin, T.-Y., et al. (2017). Focal loss for dense object detection. *ICCV*.
- Oom, D., et al. (2016). Biome-specific sensitivity of fire size distributions to the time-gap parameter. *Remote Sensing*, 8, 663.
- Pinto, M. M., et al. (2020). A deep learning approach for mapping and dating burned areas using temporal sequences of satellite images. *ISPRS J. Photogramm.*, 160.
- Shi, X., et al. (2015). Convolutional LSTM network: A machine learning approach for precipitation nowcasting. *NeurIPS*.
- Tan, C., et al. (2023). OpenSTL: A comprehensive benchmark of spatio-temporal predictive learning. *NeurIPS Datasets & Benchmarks*.
- Wooster, M. J., et al. (2015). LSA SAF Meteosat FRP products — Part 1. *Atmos. Chem. Phys.*, 15.
- Zhou, X., Wang, D., & Krähenbühl, P. (2019). Objects as points. arXiv:1904.07850.